# A COUNTEREXAMPLE TO THE POPA–SCHNELL GLOBAL GENERATION CONJECTURE

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Abstract. We provide a counterexample to the Popa–Schnell global generation conjecture. The counterexample consists of a contraction from a smooth projective threefold to a smooth projective surface.

## 1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Popa and Schnell [PS14, Conjecture 1.3] predicted that for any contraction $f : X \to Y$ between smooth projective varieties and any ample line bundle $\mathcal{L}$ on $Y$, the sheaf

$$f_*\omega_X \otimes \mathcal{L}^{\otimes(\dim Y+1)}$$

is globally generated. When $f$ is the identity morphism, this is nothing but Fujita's base point freeness conjecture. Since Fujita's base point freeness conjecture is now a theorem [OAI26], it is natural to ask whether the Popa–Schnell conjecture holds. In this paper, however, we provide a counterexample to [PS14, Conjecture 1.3].

**Theorem 1.1.** *There exist a contraction $f : X \to Y$ from a smooth projective threefold $X$ to a smooth projective surface $Y$ and an ample line bundle $\mathcal{L}$ on $Y$, such that*

$$f_*\omega_X \otimes \mathcal{L}^{\otimes 3}$$

*is not globally generated. In particular,* [PS14, Conjecture 1.3] *fails.*

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## 2. Proof of Theorem 1.1

We adopt the standard notation and terminology for the minimal model program from [Sho92, KM98, BCHM10] and use them freely. We refer the reader to [Laz04] for preliminaries on multiplier ideal sheaves. A *contraction* is a projective surjective morphism $f : X \to Y$ between normal varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Y$.

*Proof of Theorem 1.1.* We divide the construction into several steps.

**Step 1.** In this step we construct $Y$ and $\mathcal{L}$.

Consider $\mathbb{P}^2 = \mathbb{P}^2_{xyz}$ with coordinates $[x : y : z]$. Let $u := \frac{x}{z}$ and $v := \frac{y}{z}$. Write

$$C_0 := (y^3z - x^4 = 0) \subset \mathbb{P}^2, \quad o := [0 : 0 : 1].$$

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Then $C_0$ is an integral quartic curve whose only singular point is $o$. Let $Q$ be a general quartic curve on $\mathbb{P}^2$. Then every member of the pencil spanned by $C_0$ and $Q$ is integral, as the non-integral quartics form a closed subset of $|\mathcal{O}_{\mathbb{P}^2}(4)| \cong \mathbb{P}^{14}$ of dimension 11 whose join with $[C_0]$ has dimension at most 12, and

$$\{q_1, \dots, q_{16}\} := C_0 \cap Q \subset C_0 \backslash \{o\}$$

for some closed points $q_1, \dots, q_{16}$. Let $h : Y \to \mathbb{P}^2$ be the blow-up of $q_1, \dots, q_{16}$, $E_i := h^{-1}(q_i)$, $Q_Y := h_*^{-1}Q$, and $C := h_*^{-1}C_0$. Then we have

$$Q_Y = h^*Q - \sum_{i=1}^{16} E_i \quad \text{and} \quad C = h^*C_0 - \sum_{i=1}^{16} E_i.$$

Let $y_0 \in C$ be the closed point such that $h(y_0) = o$. Then $C$ and $Q_Y$ do not intersect and $C \sim Q_Y \geq 0$, so $|C|$ is a base-point-free linear system. We have that $C^2 = 0$, so $|C|$ defines a contraction $p : Y \to \mathbb{P}^1$. The fibers of $p$ are the strict transforms of the members of the pencil spanned by $C_0$ and $Q$, so every fiber of $p$ is irreducible. We let $\Gamma$ be a general fiber of $p$, $L := E_1 + 2\Gamma$, and $H$ a general element of $|h^*\mathcal{O}_{\mathbb{P}^2}(1)|$. Then we have that $C \sim \Gamma$ and $4H \sim C + \sum_{i=1}^{16} E_i$. We also have that

$$L^2 = 3, \quad L \cdot E_1 = L \cdot \Gamma = 1.$$

Let $l \neq E_1$ be an integral curve on $Y$. If $\Gamma \cdot l = 0$, then $l$ is a fiber of $p$, so $L \cdot l = 1$. Otherwise $L \cdot l = E_1 \cdot l + 2\Gamma \cdot l \geq 2$. Thus $L$ is ample by the Nakai–Moishezon criterion. We have

$$K_Y = h^*K_{\mathbb{P}^2} + \sum_{i=1}^{16} E_i \sim -3H + \sum_{i=1}^{16} E_i,$$

so

$$K_Y \cdot C = \left(-\frac{3}{4}C + \frac{1}{4}\sum_{i=1}^{16} E_i\right) \cdot C = 4.$$

Thus $g(\Gamma) = 3$. Since $\deg(L|_\Gamma) = L \cdot \Gamma = 1$, $L|_\Gamma$ is not base-point-free. Thus $L$ is not base-point-free. We let $\mathcal{L} := \mathcal{O}_Y(L)$.

**Step 2.** In this step we construct a birational morphism $g : Y' \to Y$ and a reduced divisor $B$ on $Y$ and show that $\mathcal{J}(Y, \frac{12}{13}B) = \mathfrak{m}_{y_0}^2$.

By an explicit computation, we may let $g : Y' \to Y$ be a log resolution of $(Y, C)$ near $y_0$ with exceptional divisors $D_1, \dots, D_4$ satisfying the following:

(1) $\cup_{i=1}^4 D_i$ is a chain with $-D_i^2 = 2, 2, 1, 4$ for $i = 1, 2, 3, 4$.
(2) $C' := g_*^{-1}C$ intersects $D_3$ transversely and $C'^2 = -12$.

We have that

$$K_{Y'} + cC' + (-2 + 4c)D_1 + (-4 + 8c)D_2 + (-6 + 12c)D_3 + (-1 + 3c)D_4 = g^*(K_Y + cC)$$

for any real number $c$. Let $\Gamma_1, \dots, \Gamma_{12}$ be general fibers of $p$ and let

$$B := C + \sum_{i=1}^{12} \Gamma_i \in |13\Gamma|.$$

Since $C, \Gamma_1, \dots, \Gamma_{12}$ are distinct fibers of $p$, they are pairwise disjoint, so $g$ is a log resolution of $(Y, B)$ which is an isomorphism over $Y \backslash \{y_0\}$, $(Y, \frac{12}{13}B)$ is klt outside $y_0$, and

$$g^*B = C' + \sum_{i=1}^{12} g_*^{-1}\Gamma_i + 4D_1 + 8D_2 + 12D_3 + 3D_4.$$

We have

$$\mathcal{J}\left(Y, \frac{12}{13}B\right) = g_*\mathcal{O}_{Y'}\left(K_{Y'/Y} - \left\lfloor \frac{12}{13}g^*B \right\rfloor\right) = g_*\mathcal{O}_{Y'}(-D_1 - 3D_2 - 5D_3 - D_4)$$

and

$$\mathfrak{m}_{y_0}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-D_1 - 2D_2 - 3D_3 - D_4) =: \mathcal{O}_{Y'}(-G)$$

so

$$\mathfrak{m}_{y_0}^2\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(-2G) \subset \mathcal{O}_{Y'}(-D_1 - 3D_2 - 5D_3 - D_4) \subset \mathcal{O}_{Y'}(-G) = \mathfrak{m}_{y_0}\mathcal{O}_{Y'}.$$

Thus

$$\mathfrak{m}_{y_0}^2 \subset \mathcal{J}\left(Y, \frac{12}{13}B\right) \subset g_*\mathcal{O}_{Y'}(-G) = \mathfrak{m}_{y_0}.$$

Pick any $a \in \mathfrak{m}_{y_0}\backslash\mathfrak{m}_{y_0}^2$ and let $D$ be the divisor $(a = 0)$ on a neighborhood of $y_0$. Then $g_*^{-1}D \cdot G = 1$, so

$$(g_*^{-1}D \cdot D_i)_{1\le i\le 4} \in \{(1,0,0,0),(0,0,0,1)\}.$$

The equalities $g^*D \cdot D_i = 0$ then imply that $\operatorname{mult}_{D_3} D = 3$ or 4, so $a \notin \mathcal{J}(Y, \frac{12}{13}B)$. Therefore,

$$\mathcal{J}\left(Y, \frac{12}{13}B\right) = \mathfrak{m}_{y_0}^2.$$

**Step 3.** Let $N := K_Y + 12\Gamma + 3L$. In this step we show that

$$\mathcal{O}_Y(N) \otimes \mathcal{J}\left(Y, \frac{12}{13}B\right)$$

is not globally generated at $y_0$.

Let $y_0' := C' \cap D_3$. Then we have

$$\mathfrak{m}_{y_0}\mathcal{O}_{C'} = \mathcal{O}_{C'}(-3y_0').$$

We also have $\mathcal{O}_Y(-C) \subset \mathfrak{m}_{y_0}^3$ as $C_0 = (v^3 = u^4)$ has multiplicity 3 at $o$. Let

$$\mathcal{Q} := \mathfrak{m}_{y_0}^2\mathcal{O}_C = \mathfrak{m}_{y_0}^2/\mathcal{O}_Y(-C).$$

Then

$$\mathcal{Q} \subset (g|_{C'})_*\mathcal{O}_{C'}(-6y_0')$$

and

$$N \cdot C = K_Y \cdot C + 3L \cdot C = 7,$$

so by the projection formula, we have

$$h^0(Y, \mathcal{O}_Y(N) \otimes \mathcal{Q}) \le h^0(C', \mathcal{O}_{C'}((g|_{C'})^*N - 6y_0')) = h^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(1)) = 2.$$

On the other hand, as $\mathcal{O}_Y(-C) \subset \mathfrak{m}_{y_0}^3$, the surjection $\mathfrak{m}_{y_0}^2 \to \mathcal{Q}$ induces an isomorphism

$$\mathfrak{m}_{y_0}^2 \otimes k(y_0) = \mathfrak{m}_{y_0}^2/\mathfrak{m}_{y_0}^3 \cong \mathcal{Q} \otimes k(y_0)$$

of vector spaces of dimension 3. The evaluation map of $\mathcal{O}_Y(N) \otimes \mathfrak{m}_{y_0}^2$ at $y_0$ factors through $H^0(Y, \mathcal{O}_Y(N) \otimes \mathcal{Q})$, so its rank is at most $2 < 3$. Thus $\mathcal{O}_Y(N) \otimes \mathfrak{m}_{y_0}^2 = \mathcal{O}_Y(N) \otimes \mathcal{J}\left(Y, \frac{12}{13}B\right)$ is not globally generated at $y_0$.

**Step 4.** In this step we construct a cyclic cover of $Y'$ and a variety $Z$. Write $B = (s = 0)$ for some $s \in |13\Gamma|$. Let $\pi : S \to Y'$ be the cyclic cover obtained by taking the 13-th root of $g^*s$ with the associated $\mu_{13}$-action [EV92, Section 3.5 and Corollary 3.11]. Then $S$ is an irreducible normal projective surface [EV92, Section 3.5 and Lemma 3.15(a)], $\pi$ is finite of degree 13, and $\pi$ is étale over $Y'\backslash \operatorname{Supp} g^*B$ [EV92, Lemma 3.15(b)]. Since $g^*B$ is snc, by [EV92, Lemma 3.24], $S$ has at most quotient singularities. In particular, $S$ is klt. We have

$$\pi_*\mathcal{O}_S = \bigoplus_{j=0}^{12} \mathcal{O}_{Y'}\left(-jg^*\Gamma + \left\lfloor \frac{j}{13}g^*B \right\rfloor\right)$$

(cf. [EV92, Corollary 3.11]). Since $\pi$ is finite, by duality, we have

$$\pi_*\omega_S = \bigoplus_{j=0}^{12} \mathcal{O}_{Y'}\left(K_{Y'} + jg^*\Gamma - \left\lfloor \frac{j}{13}g^*B \right\rfloor\right).$$

We let $\sigma : Z \to S$ be the minimal resolution of $S$ and let $\psi := g \circ \pi \circ \sigma : Z \to Y$. Then we have

$$\psi_*\mathcal{O}_Z = \bigoplus_{j=0}^{12} g_*\mathcal{O}_{Y'}\left(-jg^*\Gamma + \left\lfloor \frac{j}{13} g^*B \right\rfloor\right) \quad \text{and} \quad \psi_*\omega_Z = \bigoplus_{j=0}^{12} \mathcal{O}_Y(K_Y + j\Gamma) \otimes \mathcal{J}\left(Y, \frac{j}{13}B\right).$$

**Step 5.** In this step we construct a variety $T$. Let $\Sigma$ be a smooth projective curve of genus 2 and let $\eta$ be a divisor on $\Sigma$ such that $13\eta \sim 0$ but $\eta \not\sim 0$. Then there exists an index 13 cover $T \to \Sigma$ defined by $\eta$. By the Hurwitz formula, $g(T) = 14$. Moreover,

$$H^0(T, \mathcal{O}_T) = \bigoplus_{i=0}^{12} H^0(\Sigma, -i\eta) \cong \mathbb{C} \quad \text{and} \quad H^0(T, K_T) = \bigoplus_{i=0}^{12} H^0(\Sigma, K_\Sigma - i\eta).$$

In particular, $H^0(T, K_T)_{\chi^{12}} \neq 0$.

**Step 6.** In this step we construct $f : X \to Y$ and show that it is a contraction. We have constructed two actions of $\mu_{13}$: one on $S$ with quotient map $\pi : S \to Y'$, which lifts to $Z$ by the uniqueness of minimal resolution, with generator $\zeta_1$; and one on $T$ with quotient map $T \to \Sigma$, with generator $\zeta_2$. We let $\mu_{13}$ act on $Z \times T$ diagonally with generator $(\zeta_1, \zeta_2)$ and let $X := (Z \times T)/\mu_{13}$. Then the quotient map $q : Z \times T \to X$ is finite étale of degree 13 and $X$ is smooth projective as the action of $\mu_{13}$ on $Z \times T$ is free. Let $\mathrm{pr}_Z : Z \times T \to Z$ and $\mathrm{pr}_T : Z \times T \to T$ be the projections. Then $\psi \circ \mathrm{pr}_Z : Z \times T \to Y$ is $\mu_{13}$-invariant, so it induces a projective surjective morphism $f : X \to Y$. Since $q$ is étale,

$$q^*\omega_X = \omega_{Z\times T} = \mathrm{pr}_Z^*\,\omega_Z \otimes \mathrm{pr}_T^*\,\omega_T.$$

By considering the $\mu_{13}$-invariants and the Künneth formula, we have

$$f_*\mathcal{O}_X = \left(\psi_*\mathcal{O}_Z \otimes_{\mathbb{C}} H^0(T, \mathcal{O}_T)\right)^{\mu_{13}} = (\psi_*\mathcal{O}_Z)^{\mu_{13}} = \mathcal{O}_Y,$$

so $f$ is a contraction.

**Step 7.** We conclude the proof of Theorem 1.1. We have

$$f_*\omega_X = \left(\psi_*\omega_Z \otimes_{\mathbb{C}} H^0(T, \omega_T)\right)^{\mu_{13}} = \bigoplus_{j=0}^{12} (\psi_*\omega_Z)_{\chi^{-j}} \otimes_{\mathbb{C}} H^0(T, \omega_T)_{\chi^j}.$$

Since $H^0(T, \omega_T)_{\chi^{12}} \neq 0$, if $f_*\omega_X \otimes \mathcal{L}^{\otimes 3}$ is globally generated, then

$$(\psi_*\omega_Z)_{\chi^{-12}} \otimes \mathcal{O}_Y(3L)$$

is globally generated. This is not possible as

$$(\psi_*\omega_Z)_{\chi^{-12}} \otimes \mathcal{O}_Y(3L) = \mathcal{O}_Y(K_Y + 12\Gamma + 3L) \otimes \mathcal{J}\left(Y, \frac{12}{13}B\right)$$

by **Step 4**, which is not globally generated by **Step 3**. □

**Remark 2.1** (Intuition of the construction)**.** The example was found by an AI system, and its correctness is easy to verify. We briefly explain the intuition behind the construction.

First of all, [PS14, Conjecture 1.3] is actually stated for any morphism, and in this generality it has verbatim trivial counterexamples. A more general consideration is the case when $f$ is only assumed to be a projective surjective morphism. We keep this in mind, but in the following we first consider contractions only.

The key point is that the known results already show where a counterexample could be found. The conjecture [PS14, Conjecture 1.3] has been proved in the following cases:

(1) When $Y$ is an abelian variety [PS14, Corollary 5.4].
(2) When $\mathcal{L}$ is globally generated [PS14, Theorem 1.4] (see also [Eji23]). We remark that [PS14, Conjecture 1.3] fails in positive characteristic even when $\mathcal{L}$ is globally generated [SZ20] (see also [Eji24]).
(3) When $Y$ is a curve [PS14, Proposition 2.11].

(4) When $f$ is birational, in which case $f_*\omega_X = \omega_Y$ and the statement becomes Fujita's base point freeness conjecture [OAI26].

(2)–(4) above indicate that, to find a counterexample, we need $\mathcal{L}$ to be not globally generated and $\dim X > \dim Y \geq 2$. In our case, $\dim X = 3$ and $\dim Y = 2$, which are the minimal dimensions where a counterexample could be found.

So let us try to find a counterexample when $\dim Y = 2$. In this case, we have:

(5) $f_*\omega_X \otimes \mathcal{L}^{\otimes 3}$ is globally generated at a general point $y \in Y$ ([Dut20, Corollary B], [DM19, Remark 5.2]).

(6) $f_*\omega_X \otimes \mathcal{L}^{\otimes 3}$ is globally generated at any point $y$ over which $f$ is smooth and $\omega_X$ is relatively free ([Dut20, Theorem C]).

(7) $f_*\omega_X \otimes \mathcal{L}^{\otimes l}$ is globally generated at every regular value $y$ of $f$ if

$$l \geq \left\lfloor \frac{2}{\varepsilon(\mathcal{L}, y)} \right\rfloor + 1,$$

where $\varepsilon(\mathcal{L}, y)$ is the Seshadri constant of $\mathcal{L}$ at $y$ ([Den21, Theorem A], see also [Iwa20]).

In short, to find a counterexample when $\dim Y = 2$, one should look for a fibration over a surface which is singular over a point where an ample but not globally generated line bundle has a small Seshadri constant. This exactly matches Miranda's construction [EL93, Example 3.1, p. 182]: there exist a smooth projective surface $Y$ with a fibration $p : Y \to \mathbb{P}^1$ which has a rational cuspidal quartic fiber $C$, and an ample line bundle $\mathcal{L}$ on $Y$ with $\mathcal{L} \cdot C = 1$, so $\varepsilon(\mathcal{L}, y_0) \leq \frac{1}{3}$ at the cusp $y_0$ of $C$. In particular, $\mathcal{L}$ is not globally generated.

Now the idea is to realize a direct summand of $f_*\omega_X \otimes \mathcal{L}^{\otimes 3}$ as

$$\mathcal{O}_Y(N) \otimes \mathcal{J}, \quad N := K_Y + \lambda C + 3L,$$

for some positive integer $\lambda$ and some multiplier ideal sheaf $\mathcal{J}$. The degree of $\mathcal{O}_Y(N)$ along $C$ is $N \cdot C = 7$, independent of $\lambda$, while $C$ has multiplicity 3 at $y_0$: the sections of $\mathcal{O}_Y(N) \otimes \mathfrak{m}_{y_0}^2$ restrict to sections of degree $7 - 6 = 1$ on the normalization $\mathbb{P}^1$ of $C$, which form a space of dimension 2, whereas $\mathfrak{m}_{y_0}^2/\mathfrak{m}_{y_0}^3$ has dimension 3. Hence

$$\mathcal{O}_Y(N) \otimes \mathfrak{m}_{y_0}^2$$

is not globally generated at $y_0$, and we want $\mathcal{J} = \mathfrak{m}_{y_0}^2$.

Now we consider the projective surjective morphism version of [PS14, Conjecture 1.3] and construct a counterexample $\psi : Z \to Y$ that is generically finite. The idea is to take $\psi : Z \to Y$ to be a resolution of the cyclic cover of degree $\lambda+1$ branched along a reduced divisor $B \in |(\lambda+1)C|$, so that

$$\psi_*\omega_Z = \bigoplus_{j=0}^{\lambda} \mathcal{O}_Y(K_Y + jC) \otimes \mathcal{J}\left(Y, \frac{j}{\lambda+1}B\right).$$

As long as one of the summands $\mathcal{O}_Y(K_Y + jC) \otimes \mathcal{J}\left(Y, \frac{j}{\lambda+1}B\right) \otimes \mathcal{L}^{\otimes 3}$ is not globally generated, $\psi_*\omega_Z \otimes \mathcal{L}^{\otimes 3}$ is not globally generated. So all we need is to take $j = \lambda$ and require that $\mathcal{J}\left(Y, \frac{\lambda}{\lambda+1}B\right) = \mathfrak{m}_{y_0}^2$. Since $|C|$ is base point free, we may take $B = C + \sum_{i=1}^{\lambda} \Gamma_i$ with $\Gamma_1, \ldots, \Gamma_\lambda$ general members of $|C|$, that is, general fibers of $p$. An explicit computation shows that $\mathcal{J}\left(Y, \frac{\lambda}{\lambda+1}B\right) = \mathfrak{m}_{y_0}^2$ whenever $\lambda \geq 11$. We take $\lambda = 12$ instead of 11, as $13 = \lambda + 1$ is a prime number and the arguments on cyclic covers of degree $\lambda + 1$ can then be stated more easily. In practice, $Z$ is constructed by first taking a log resolution $Y'$ of $(Y, C)$, then a cyclic cover $S \to Y'$ of degree 13, and then the minimal resolution $Z$ of $S$.

Finally, to obtain a counterexample in the contraction case, all we need to do is to multiply $Z$ by a curve $T$ with a free action of $\mu_{13}$ and take the quotient $X := (Z \times T)/\mu_{13}$, as the summand structure is preserved by the Künneth formula. We take $T$ to be an étale cyclic cover of degree 13 of a curve $\Sigma$ of genus 2, defined by a nontrivial 13-torsion divisor $\eta$ on $\Sigma$; by the Hurwitz

formula, $g(T) = 14$. Here we need $g(\Sigma) \geq 2$ to ensure that the summand survives, which in turn relies on the existence of sections in $H^0(\Sigma, \omega_\Sigma - i\eta)$ for $1 \leq i \leq 12$: by the Riemann–Roch theorem, $h^0(\Sigma, \omega_\Sigma - i\eta) = g(\Sigma) - 1$.

## References


[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. M$^{\mathrm{c}}$Kernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[Den21] Y. Deng, *Applications of the Ohsawa–Takegoshi extension theorem to direct image problems*, Int. Math. Res. Not. **2021** (2021), no. 23, 17611–17633.

[Dut20] Y. Dutta, *On the effective freeness of the direct images of pluricanonical bundles*, Ann. Inst. Fourier (Grenoble) **70** (2020), no. 4, 1545–1561.

[DM19] Y. Dutta and T. Murayama, *Effective generation and twisted weak positivity of direct images*, Algebra Number Theory **13** (2019), no. 2, 425–454.

[EL93] L. Ein and R. Lazarsfeld, *Seshadri constants on smooth surfaces*, Astérisque **218** (1993), 177–186.

[Eji23] S. Ejiri, *Notes on direct images of pluricanonical bundles*, Eur. J. Math. **9** (2023), no. 4, Paper No. 96.

[Eji24] S. Ejiri, *Direct images of pluricanonical bundles and Frobenius stable canonical rings of fibers*, Algebr. Geom. **11** (2024), no. 1, 71–110.

[EV92] H. Esnault and E. Viehweg, *Lectures on Vanishing Theorems*, DMV Seminar **20**, Birkhäuser, Basel, 1992.

[Iwa20] M. Iwai, *On the global generation of direct images of pluri-adjoint line bundles*, Math. Z. **294** (2020), no. 1-2, 201–208.

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge, 1998.

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. II*, Ergeb. Math. Grenzgeb. (3) **49**, Springer-Verlag, Berlin, 2004.

[OAI26] OpenAI, *Fujita's freeness conjecture*, https://github.com/openai/math.

[PS14] M. Popa and C. Schnell, *On direct images of pluricanonical bundles*, Algebra Number Theory **8** (2014), no. 9, 2273–2295.

[SZ20] J. Shentu and Y. Zhang, *On the simultaneous generation of jets of the adjoint bundles*, J. Algebra **555** (2020), 52–68.

[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.



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